

Maths Homework Template

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May 8, 2022

1. Answer to question 1:

Use inline equations for simple math $1 + 1 = 2$, and centered equations for more involved or important equations:

$$a^2 + b^2 = c^2. \quad (1)$$

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \quad (2)$$

It is recommended to write scalars without boldface like $E = \hbar\omega$, vectors in boldface, and matrices in sans-serif:

$$\boldsymbol{p} = \hbar \boldsymbol{k} \quad (3)$$

$$\frac{\|\mathbf{A}\mathbf{x}\|_\infty}{\|\mathbf{x}\|_\infty} = \frac{\max_i \left| \sum_{j=1}^n \mathbf{A}_{ij} x_{j1} \right|}{\max_i |x_{i1}|} \quad (4)$$

¶ An example of a matrix:

$$\mathbf{M} = \begin{bmatrix} \frac{1+\sqrt{3}}{2} & e^{-i\omega t} \\ -e^{-i\omega t} & \frac{1-\sqrt{3}}{2} \end{bmatrix} \quad (5)$$

¶ An example of intergals:

$$\iiint \nabla \cdot \boldsymbol{\Phi} \, dV = \oiint \boldsymbol{\Phi} \, d\mathbf{S} \quad (6)$$

¶ An example of roots:

$$r = \sqrt[3]{\frac{3}{4\pi N}} \quad (7)$$

2. Answer to question 2:

$$\begin{aligned} I(x) &= \int_{-\frac{b}{2}}^{\frac{b}{2}} dI(x) \\ &= 4\alpha \frac{I_0}{b} \int_{-\frac{b}{2}}^{\frac{b}{2}} \cos^2 \left(\frac{\pi d}{\lambda D} \left(x + \frac{D}{R} \xi \right) \right) d\xi \\ &= 4\alpha I_0 \left(1 + \frac{1}{u} \sin u \cos \left(\frac{2\pi d}{\lambda D} x \right) \right), \quad u = \frac{\pi db}{\lambda R} \end{aligned}$$

3. Answer to question 3:

$$Y_n(y) = \begin{cases} 0, & n = 2k \\ \frac{8a^2}{\pi^3 n^3} \left(1 - \frac{\cosh \frac{n\pi y}{a}}{\cosh \frac{n\pi b}{2a}} \right), & n = 2k + 1 \end{cases}$$

4. Answer to question 4:

$$\begin{aligned} u(x, t) &= \frac{6Al(-1)^m}{ES(2m+1)^2\pi^2} \sin \frac{2m+1}{2l}\pi x \sin \frac{2m+1}{2l}\pi at \\ &+ \frac{2A(-1)^{m+1}}{ES(2m+1)\pi} \left(x \cos \frac{2m+1}{2l}\pi x \sin \frac{2m+1}{2l}\pi at + at \sin \frac{2m+1}{2l}\pi x \cos \frac{2m+1}{2l}\pi at \right) \\ &+ \frac{8Al(2m+1)}{ES\pi^2} \sum_{k=0, k \neq m}^{\infty} \frac{(-1)^k}{(2k+1)[(2m+1)^2 - (2k+1)^2]} \sin \frac{2k+1}{2l}\pi x \sin \frac{2k+1}{2l}\pi at \end{aligned}$$

5. Answer to question 5:

$$\begin{aligned} u(x, t) &= \sum_{n=0}^{\infty} X_n(x) Y_n(y) \\ &= \frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} \left(1 - \frac{\cosh \frac{2k+1}{a}\pi y}{\cosh \frac{2k+1}{2a}\pi b} \right) \sin \frac{2k+1}{a}\pi x \end{aligned}$$

6. Answer to question 6:

$$\begin{aligned} \mathcal{L} \left\{ \frac{1 - \cos \omega t}{t} \right\} &= \int_p^{\infty} \left(\frac{1}{p} - \frac{1}{2(p + i\omega)} - \frac{1}{2(p - i\omega)} \right) dt \\ &= \ln p - \frac{1}{2} \ln(p + i\omega) - \frac{1}{2} \ln(p - i\omega) \Big|_p^{\infty} \\ &= \frac{1}{2} \ln \frac{p^2}{p^2 + \omega^2} \Big|_p^{\infty} \\ &= \frac{1}{2} \ln \frac{p^2 + \omega^2}{p^2} \end{aligned}$$

7. Answer to question 7:

$$\begin{aligned} [q, p]_{Q,P} &= \frac{\partial q}{\partial Q} \cdot \frac{\partial p}{\partial P} - \frac{\partial q}{\partial P} \cdot \frac{\partial p}{\partial Q} \\ &= \left(-\frac{\sqrt{1 - P^2 e^{2Q}}}{e^Q} - \frac{P^2 e^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) \cdot \left(\frac{-e^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) - \left(\frac{-Pe^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) \cdot \left(\frac{-Pe^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) \\ &= 1 + \frac{P^2 e^{2Q}}{1 - P^2 e^{2Q}} - \frac{P^2 e^{2Q}}{1 - P^2 e^{2Q}} \\ &= 1 \end{aligned}$$

8. Answer to question 8:

$$\begin{aligned} \frac{4\pi R_{\odot}^2 \cdot \sigma T_{\odot}^4}{4\pi d^2} &= 1352.83 \text{ W/m}^2 \\ T_{\odot} &= 5763.43 \text{ K} \\ \lambda_{\odot} &= \frac{b}{T_{\odot}} = 5027.86 \text{ \AA} \end{aligned}$$